

Midpoint preservation on a quarter ball

`MathlibAnnex.IsometryEquiv.map_midpoint_of_mem_ball`

Obtains a uniform local midpoint law from symmetric-lens center rigidity.

Statement

Let $f:s \rightarrow t$ and let $c \in s$. Assume $R > 0$, $\text{ball}(c,R) \subseteq s$, and $\text{ball}(f(c),R) \subseteq t$. If $x,y \in s$ both lie in $\text{ball}(c,R/4)$, then f sends their midpoint to the midpoint of $f(x)$ and $f(y)$.

Assumptions

Real normed vector spaces; radius- R balls about c and $f(c)$ lie in the source and target; $R > 0$; x,y lie in the radius- $R/4$ source ball.

Conclusion

f sends the midpoint of x and y to the midpoint of $f(x)$ and $f(y)$.

Proof idea

The two endpoints are less than $R/2$ apart. Their symmetric lens of radius $R/2$ lies in the source R -ball, and the corresponding target lens lies in the target R -ball. The subset-local lens theorem therefore applies.

Proof steps

1. Use the triangle inequality through c to bound $\text{dist}(x,y)$ by less than $R/2$.
2. Deduce the half-distance condition needed for a symmetric lens of radius $R/2$.
3. Show every point of the source lens lies in $\text{ball}(c,R)$ by combining its distance to x with the $R/4$ bound on x .
4. Transport the $R/4$ bound from x to $f(x)$, then prove the analogous containment of the target lens in $\text{ball}(f(c),R)$.
5. Apply midpoint preservation for a set isometry whose corresponding lenses remain inside the source and target sets.

Main citations

`MathlibAnnex.IsometryEquiv.map_midpoint_of_symmetricLens_subset`

The proof or construction of Midpoint preservation on a quarter ball uses the project declaration “Midpoint preservation under lens containment” at the indicated step.

`MathlibAnnex.symmetricLens`

The proof or construction of Midpoint preservation on a quarter ball uses the project declaration “Symmetric lens” at the indicated step.

Lean source signature (exact)

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theorem map_midpoint_of_mem_ball
  {s : Set E} {t : Set F} (f : s  $\approx_i$  t) (c : s) {R :  $\mathbb{R}$ } (hR : 0 < R)
  (hsball : ball (c : E) R  $\subseteq$  s)
  (htball : ball ((f c : t) : F) R  $\subseteq$  t)
  (x y : s) (hx : (x : E)  $\in$  ball (c : E) (R / 4))
  (hy : (y : E)  $\in$  ball (c : E) (R / 4)) :
  ((f (midpoint  $\mathbb{R}$  (x : E) (y : E)),
  hsball (by
  rw [mem_ball]
  calc
  dist (midpoint  $\mathbb{R}$  (x : E) (y : E)) (c : E)  $\leq$ 
  (dist (x : E) (c : E) + dist (y : E) (c : E)) / 2 := by
  simp [dist_comm] using
  dist_midpoint_midpoint_le (x : E) (y : E) (c : E) (c : E)
  _ < R := by rw [mem_ball] at hx hy; linarith)) : t) : F) =
  midpoint  $\mathbb{R}$  ((f x : t) : F) ((f y : t) : F)
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Read exact MathlibAnnex v0.4.0 source

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AI-assisted material. Selected, edited, and reviewed by Ryotaro Tanaka. The recorded review is source-exposition correspondence, not an independent proof audit.